Project Proposal

Badminton Stroke Classification using AI Computer Vision

(Contribution to long-term Badminton Objective Player Grading Project)

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Project Overview

This project implements two distinct Machine Learning (ML) / Deep Learning (DL) models for classifying badminton strokes from video footage, tested on the ShuttleSet dataset (Wang et al., 2023). We compare accuracy and other metrics across models and benchmark against earlier results (Arthur et al., 2026).

Hunter Badminton Association (HBA) in Newcastle, NSW, hosts events for players of all skill levels. Player grading—dividing players into skill-based tiers—is decided by sub-committee, but disputes commonly arise. HBA engaged UNE to develop a more objective, AI-based grading approach. This project will build a foundation module for reliable stroke classification assisted by computer vision and time-series modelling. An interactive client portal will also be built to demonstrate the stroke classifier in action, allow classification of user-submitted videos, and display transparent metrics explaining model classifications (explainable AI).

Objectives

1. Build on the 2025 UNE Badminton Shot Classifier by developing and implementing different classification models. The conceptual framework exists, but the 2025 model's categorisations are unreliable.
2. Test models on the ShuttleSet dataset and refine.
3. Compare performance across our models, and against the previous project and published literature.
4. Provide an intuitive interface for HBA to use the system.

Client and System Requirements

Stakeholders: Academic supervisors (direct); future UNE students (long-term badminton grading project), HBA, and badminton players/coaches (indirect). See **Appendix B** for detail.

Functional Requirements

Full requirements in **Appendix C**. Summary:

Web Interface:

- Select or upload match videos [requires host/bandwidth research for videos]
- Apply metadata: draw court boundaries, identify target player, and select time-frame of stroke to categorise
- Select one or both classification models
- Submit for analysis
- Progress monitoring display
- Display classification results with transparent decision metrics (e.g. Precision thresholds, visual heat-maps).

Data Pipeline (model fit):

- Bulk video download, indexing and labelling.
- Tabular data read-in (labels and metadata)
- Feature engineering and normalisation, incl class collapse.
- Datastructures for robust data retrieval, modification and analysis

Two AI Architectures:

Explore performance gap between post-hoc and real-time models.

Target accuracy (best model): 80%+.

Note: Some papers claim 95%+, but best verifiable results are BST @ 80-85% (Chang, 2025).

Model A: spatio-temporal, resource-intensive, targeting best-in-class performance.

- Object detection -> 3D-CNN + ShuttleTracker -> Deep 1D-CNN or transformer. If transformer, possible TCN pre-processing per *BST* approach

Model B: keypoint-based, lightweight, real-time (or near) performance

- Object detection -> Keypoint graph extraction (MediaPipe) + ShuttleTracker -> TCN

Evaluation results and explainability outputs to be stored in database.

- Likely just structured JSON I/O due to scale needs vs implementation complexity.

Deployment:

Containerised for local or cloud deployment.

All tests must be reproducible using shared ShuttleSet validation data, control seeding, and consistent train/test/validation splits.

Non-Functional Requirements

Single repository containing all training and evaluation artefacts with documented training methodology. Generalised performance limitations and ecological validity constraints (inherent in ShuttleSet) must be acknowledged and addressed in evaluation design and documentation. Class taxonomy explanation. Accessibility features to be determined by DevOps/Interface team.

Project Scope

Inclusions: Web interface development, AI/ML pipeline architecture design and implementation, model training, model integration with web front end, user documentation, evaluation of model performance for shot classification.

Exclusions: Deployment hardware (not team's responsibility), production-grade scalability (proof-of-concept only), advanced player assessment capabilities (movement-pattern analysis, rally detection, player performance tracking), new annotated datasets (ShuttleSet only).

Deliverables: Stroke classification system (web interface + pipeline + two models + containerised deployment + docs), model training pipeline and artefacts, final report, presentation, handover document. See **Appendix D** for breakdown.

Feasibility and Resources

Technologies:

Computer vision for analysing footage (player movement, stroke, shuttlecock tracking). Supervised time-series deep learning on labelled gameplay videos.

Python for AI/ML; PyTorch, numpy, pandas and scikit-learn libraries; Python/JavaScript for interface. Docker for containerisation. VS Code/PyCharm. ShuttleSet Badminton Dataset. GitLab or GitHub for version control and task tracking.

GPU via UNE-HPC (access permission required; multiple systems available).

Skills:

All members competent in Python and version control. ML/CV experience varies (see **Appendix E**). Key gap: limited DL pipeline experience (one member only).

Constraints:

Dataset uses professional players (may affect generalisability). Limited advanced AI experience.

GPU access—reservations possible?

9-week timeline, currently in active research/early deliverables definition stage; delivery by 31 May 2026.

Solution Approach

A web-based interface allows users to upload badminton match footage and receive stroke classifications from selected AI models. The interface is a simple interaction layer, not production-grade software. We will review and extend last year's project repository (GitHub), improving the web interface, re-architecting where needed.

Models will be built from scratch.

Substantial data pipeline code has been shared by the author of BST; it is expected we can fork and build off this.

Specific explainable AI metrics not yet decided; however, several team members are proficient data scientists.

Alternative Approaches

Desktop Application: Viable but bloated by local AI model packaging; hardware constraints may compromise performance; less scalable than web.

CLI: Meets functional requirements but lacks appropriate UI/UX for presenting to HBA stakeholders.

Justification

Building on the existing web interface enables rapid integration of additional models and further polish before presenting to HBA.

Models require complete re-architecture due to poor predecessor performance.

Project Planning

Milestone	Due
Project proposal — final	1 April 2026
Software setup	3 April 2026
Initial prototyping sprint—task breakdown and assignments	5 April 2026
Data pipeline defined and substantially implemented. Model architectures finalised and implementation code underway.	19 April 2026
Initial model fit and evaluation attempted (both models) Database of model results implemented	26 April 2026
Hyperparameter optimisation search. Models reiterated. Front-end integration POC. Interface prototype complete	3 May 2026
Model completion final deadline Performance documentation underway Front-end MVP complete, including integration	10 May 2026
Final deadline for full non-MVP front-end, incl. final docker deployment Performance documentation complete Final report underway	17 May 2026
Final report - draft Video scripts - draft	22 May 2026
Presentation recorded	24 May 2026
Post-production editing	29 May 2026
Final report	31 May 2026

Gantt chart in **Appendix A**.

Team Member Roles

Member	Role
Ethan McDonough	Interface / UI/UX
Jared Pitman	Interface / DevOps
Kiri Lefebvre	Technical Writer / UI/UX
Isiah Darcy	ML Developer
Scott Bailey	ML Developer
Ariel Halperin	ML Developer
Curtis Martin	ML Developer

Role descriptions in **Appendix F**. Testing and documentation shared across team.

Risk Assessment

Full risk table in **Appendix G**. Key risks:

- Dataset quality issues (High/High) — previous group had significant data problems. Mitigate with early EDA and flagging issues to Arthur.
- ML/CV inexperience (High/High) — team is up-skilling. Mitigate by selecting proven architectures from literature review.
- Poor model performance (Medium/High) — two diverse algorithms maximise likelihood of acceptable results. Target 80-85% (large model) and 70-80% (small model).
- Scope creep (Medium/Medium) — this proposal is the baseline; consult Arthur before changes. Prioritise AI classification over interface if time-constrained.
- Team availability (Medium/Medium) — all members balancing other commitments. Weekly check-ins and clear task ownership mitigate.
- Domain skill gaps (Medium/Medium) — badminton knowledge low and not embedded in ML stream. Involve domain-knowledgeable members in pipeline discussions.
- Timeline overrun (Low/High) — enforce scope, defer non-essentials, run AI and web development in parallel.

References

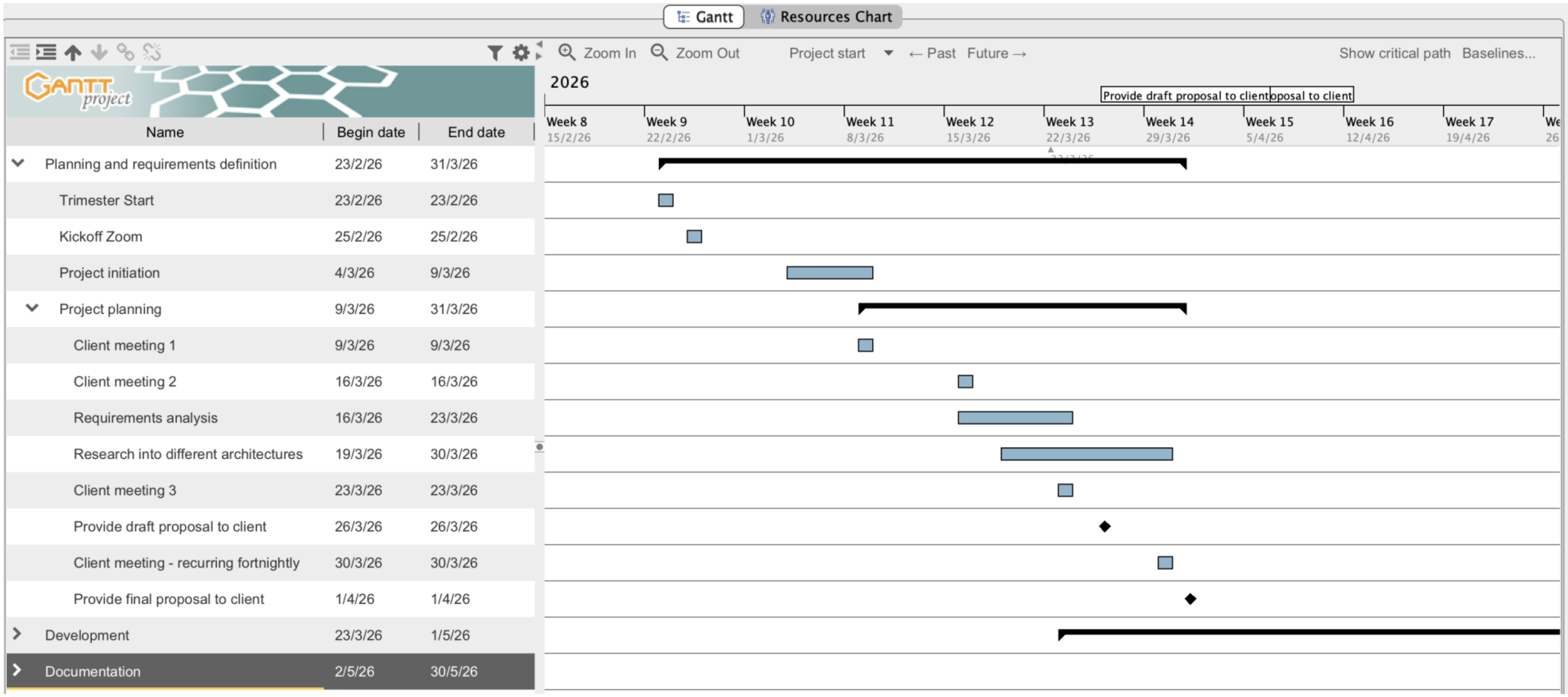
Arthur, C., Chiong, R., Ud Din, F., Hajati, F., Chakraborty, S., Paul, D., Welch, M., Crowther, R. (2026). Making the world's fastest racket sport even better: A systematic review of AI-based objective player performance assessment in badminton.

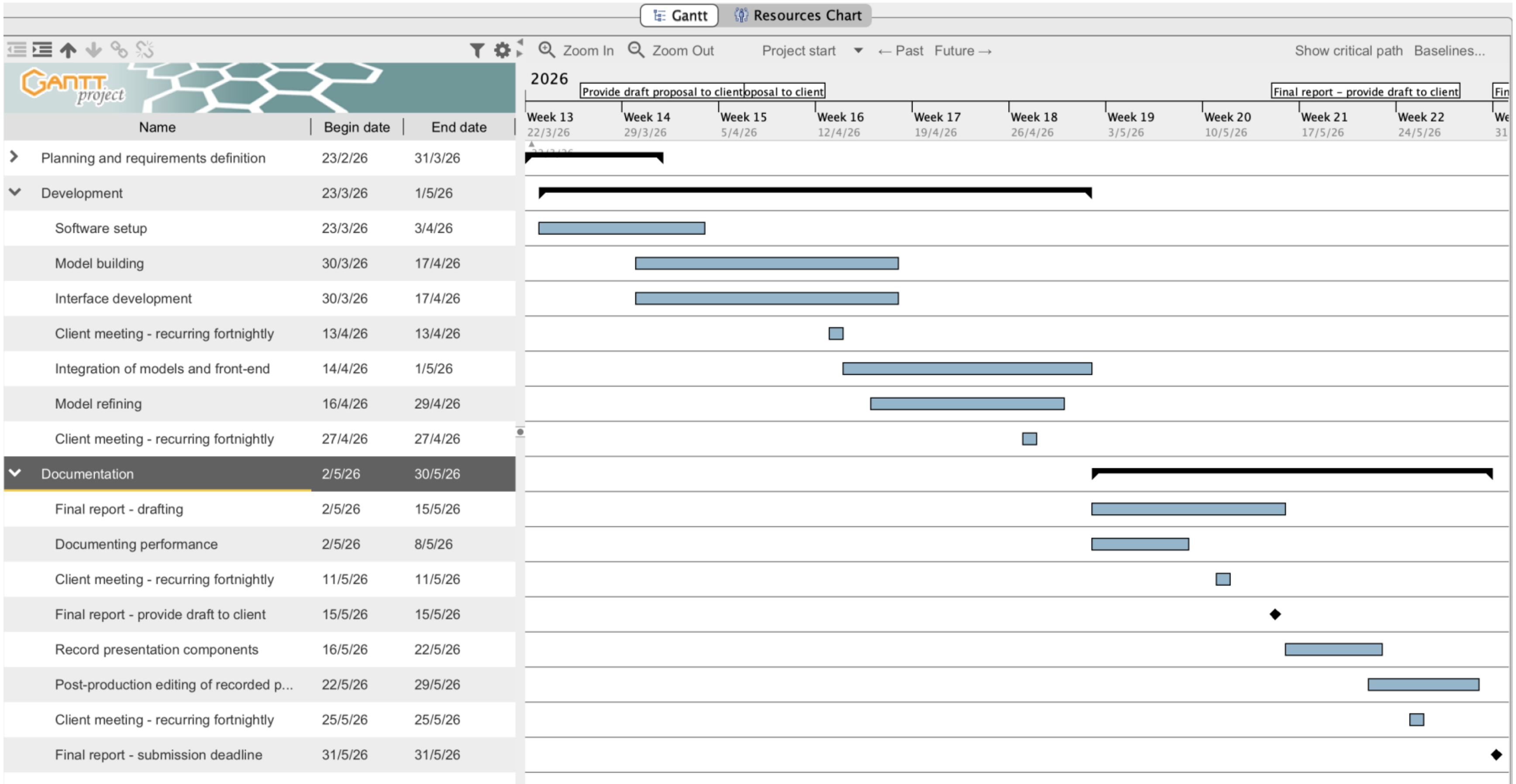
Wang, W.-Y., Huang, Y.-C., Ik, T.-U., Peng, W.-C. (2023). ShuttleSet: A human-annotated stroke-level singles dataset for badminton tactical analysis. *KDD '23*. https://arxiv.org/abs/2306.04948

Chang, J.-Y. (2025). BST: Badminton stroke-type transformer for skeleton-based action recognition in racket sports. arXiv. https://doi.org/10.48550/arxiv.2502.21085

Appendices

Appendix A — Gantt Chart





Name	Begin date	End date
Planning and requirements definition	23/2/26	31/3/26
Development	23/3/26	1/5/26
Software setup	23/3/26	3/4/26
Model building	30/3/26	17/4/26
Interface development	30/3/26	17/4/26
Client meeting - recurring fortnightly	13/4/26	13/4/26
Integration of models and front-end	14/4/26	1/5/26
Model refining	16/4/26	29/4/26
Client meeting - recurring fortnightly	27/4/26	27/4/26
Documentation	2/5/26	30/5/26
Final report - drafting	2/5/26	15/5/26
Documenting performance	2/5/26	8/5/26
Client meeting - recurring fortnightly	11/5/26	11/5/26
Final report - provide draft to client	15/5/26	15/5/26
Record presentation components	16/5/26	22/5/26
Post-production editing of recorded p...	22/5/26	29/5/26
Client meeting - recurring fortnightly	25/5/26	25/5/26
Final report - submission deadline	31/5/26	31/5/26

Appendix B — Stakeholder Details

Academic Supervisors (Arthur, Raymond, Farshid): Direct relationship. Informed project requirements, provided domain guidance from literature review. Will monitor progress and evaluate the system as part of capstone assessment.

Future Researchers (COSC320/594/595 students): Indirect. May build on outputs to extend models towards generalised player grading.

Past Researchers (COSC320/594/595 students): Indirect. Made prior contributions to stroke identification research that this project extends.

Hunter Badminton Association: Indirect. Not using the system directly, but their need for objective AI-based player assessment motivates this work. Stroke identification models may lay foundations for higher-order player performance classification.

Badminton Players & Coaches: Indirect. Potential downstream beneficiaries of improved classification models.

Appendix C — Full Functional Requirements

FR1. Web Interface

- Select match video(s) from a ShuttleSet validation set as input.
- Upload annotated videos to the validation set
- Upload unannotated videos for classification without validation metrics.
- Set parameters to tune and evaluate model performance.
- Perform markup on input video to ensure consistent model performance across varied inputs.
- Select one, two, or both classification models for the current task.
- Submit video with markup for analysis (async); display progress and updates until complete.
- Review and interpret results: stroke classifications alongside relevant frames, classification metrics, aggregated results, comparisons against previous tasks and baselines and shot distributions.

FR2. AI/ML Classification Pipelines

- Receive classification tasks from the web interface with associated metadata and video reference.
- Model A: spatio-temporal, resource-intensive, targeting best-in-class classification performance.
- Model B: spatial-only, lightweight, demonstrating the gap between computationally expensive and inexpensive approaches.
- Apply preprocessing, feature extraction, inference, and post-processing per pipeline architecture.
- Store classification results in JSON accessible to the web interface.
- Support comparative analysis between models and against literature baselines
- Models must be persistent and reproducible from training state.
- Target classification accuracy of 90–95%, consistent with literature benchmarks.

FR3. Deployment & Reproducibility

- Deployable locally via container orchestration or to cloud.

- Tests must be consistent and repeatable: shared ShuttleSet validation set, control seeding, consistent train/test/validation splits.

Non-Functional Requirements

- Single repository containing all training and evaluation artefacts.
- Training methodology (exploratory analysis, hyperparameters, code, instructions, evaluations) documented in repository.
- Accessibility features to be determined by DevOps/Interface team.
- Generalised performance limitations acknowledged and addressed through evaluation design and documentation.
- Ecological validity constraints inherent in ShuttleSet acknowledged and addressed through evaluation design and documentation.

Appendix D — Deliverables Breakdown

1. Badminton Shot Classification System

- Web interface for task creation and result review
- Two architecturally distinct shot classification pipelines trained on badminton match footage
- Containerised architecture for local and cloud deployment
- User documentation for setup and use

2. Model Training Pipeline and Artefacts

- Exploratory analysis, definitions and parameter configurations, pipeline code, model evaluations and analysis, deployment instructions

3. Final Report

- Project outcomes, system description, project diary, individual contributions, requirements/deliverables, bug report, quality testing report, lessons learned

4. Presentation

- Group/client/project overview, objectives/plans/execution/outcomes, system demo, project closure

5. Handover Document

- Objectives, architecture, outcomes, resources for reproducibility and further development

Appendix E — Team Skill Matrix

Key: w = weak, c = comfortable with concepts, ✓ = competent, x = no experience

Skill	Ethan	Isiah	Jared	Kiri	Scott	Ari	Curtis
Python	✓	✓	✓	✓	✓	✓	✓
TensorFlow	w	w	w	w	x	c	x
PyTorch	c	c	c	w	w	x	x
Data Science Libs	w	c	w	c	✓	c	x
Computer Vision	w	w	w	w	x	c	x
Timeseries (DL)	x	w	w	w	x	✓	x
Timeseries (ML/stats)	x	c	x	w	✓	x	x
Feature Engineering	w	w	c	w	✓	✓	x
Model Train/Eval	x	c	c	c	✓	✓	x
Version Control	✓	✓	✓	✓	✓	✓	✓
CI/CD & Containers	w	x	✓	c	c	x	c
Testing	✓	w	✓	✓	w	w	✓
Databases/SQL	c	✓	✓	c	✓	w	✓
Full-stack Web	c	x	✓	c	c	w	✓
Backend-GUI	w	x	✓	✓	c	✓	✓
Badminton Knowledge	c	x	c	w	w	x	x
Project Management	c	c	c	c	c	w	c

Appendix F — Role Descriptions

ML Developer: Develops and evaluates ML models, including training, testing, and optimisation.

Interface Developer: Implements the frontend and ensures it communicates with ML models.

UI/UX Developer: Designs user experience (layout, navigation, usability). Expands on Interface Developer work.

Technical Writer: Creates documentation for users and developers.

DevOps: Ensures the project runs reliably across platforms with minimal setup.

Appendix G — Full Risk Assessment

Risk	Description	L / I	Mitigation
Dataset Quality	Previous group had significant dataset issues. ShuttleSet may contain similar problems.	High/High	Conduct early EDA. Review data collection methodology. Flag quality issues (missing values, class imbalance, labelling errors) to Arthur immediately.
ML/CV Inexperience	Team is up-skilling; only one member has built a full DL timeseries pipeline. Risk of infeasible architectural decisions.	High/High	Use literature review to identify proven architectures. Prioritise well-documented designs. Produce version-controlled, incrementally developed code with reproducible training.

Poor Model Performance	Previous project achieved poor accuracy. New selections may also underperform on professional-player data.	Med/High	Set accuracy target. Implement two diverse algorithms. Use literature to guide selection toward demonstrated badminton stroke classification performance. Target performance: 80-85% (large model) and 70-80% (small model).
Scope Creep	Two workstreams (AI + web) may expand beyond agreement.	Med/Med	This proposal is baseline. Consult Arthur before changes. Prioritise AI classification over interface if time-constrained.
Team Availability	All members balancing other study/work.	Med/Med	Weekly check-ins, Slack updates, clear task ownership. Escalate to Raymond early if needed.
Technical Skill Gaps	Badminton domain knowledge low and not embedded in ML stream.	Med/Med	Involve domain-knowledgeable members in ML discussions. Seek Raymond's input on domain alignment.
Timeline Overrun	~9 weeks for two pipelines, evaluation, web interface, report, and presentation while up-skilling.	Low/High	Enforce scope. Defer non-essentials. Run AI and web development in parallel, tracking against milestones weekly.